29/7

Data and Applications

Instructor : #PKGiri

→ **Data :**

Known facts that can be recorded and have an implicit meaning

Database :

Collection of related data

DBMS :

DB + Software

Mini-world :

Some part of the real world about which data is stored in DB.

6/10

Triggers and Stored Procedure → Not part of syllabus

Keywords : Entity , Attribute , Relations

- ACID
  - Durability
  - Isolation
  - Consistency
  - Atomicity

- Constraints , Operations , Datatypes

  Database schema → Description | Catalogue

  State → Snapshot

- Three - State Architechture

  Two - tier Client - Server architechture

- DDL ← Data creation

  DML ← Query Language

→ Create the following for an event of Infinium :

(a) Tables
(b) Columns
(c) Datatypes
(d) Relationships

} Database Schema

Database schema :

Infinium → Mini-world

e.g. Euphony :

STUDENTs form a band with other STUDENTs.

BAND has multiple roles – Singer, Pianist, Guitarist, Drummer, etc.

BAND gives auditions and proceed further if selected.

CLUB MEMBERS judge the auditions. They are also STUDENTs.

BANDs performs SONGs.

BANDS are given positions based on their performance

Band ID	Performer ID
int	int

Performer ID	Talent
int	varchar

Band ID	Selected / NOT
int	varchar

Band ID	Song ID
int	int

Band ID	Rank
int	int

Song ID	Song name
int	varchar

1 Band → n Performers

1 Band → n songs

1 Performer → 1 Talent

– 2025121009

9/10

→ **ER Model**

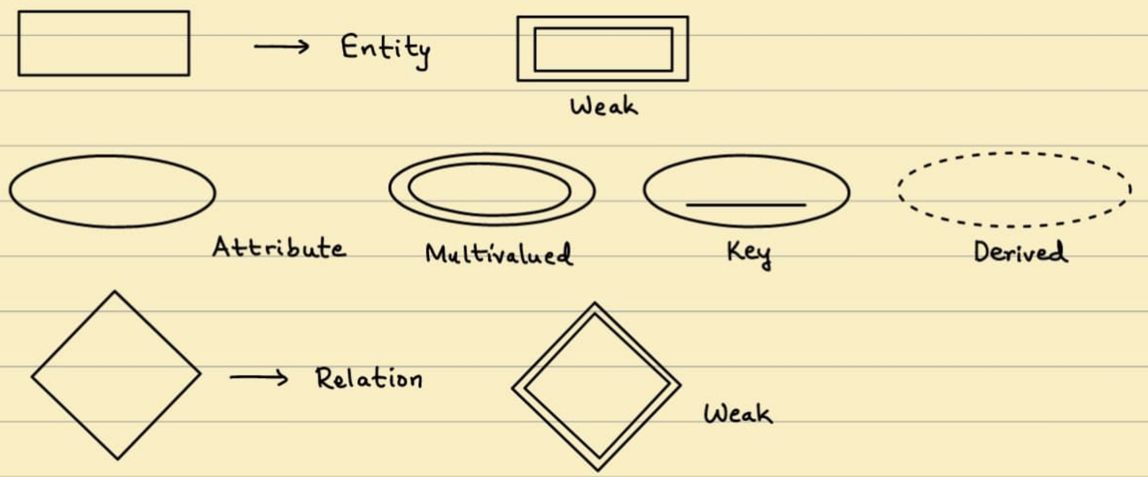
- Database Designer : doesn't care how the data is captured.
- Entity : Basic concept of ER Model (Row)

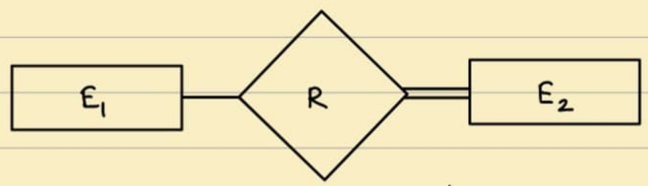
  Attributes : Properties of entity (column)

Attributes
- Simple (Atomic)
- Composite
- Multivalued

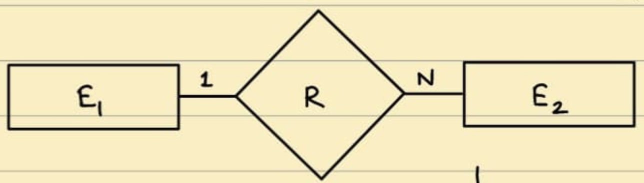
Key Attribute → Each entity has a unique value

Representation : Entity. Attribute

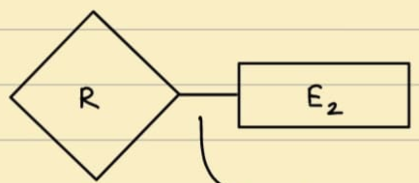




→ Total participation of E_2 in R



→ Cardinality Ratio → 1:N for $E_1 : E_2$ in R



→ (min – max) → Structural constraint

13/10

Relation types

Constraints

Recursive relations

16/10

- Weak Entity → Depends on strong Entity

- Attributes of Relationship types

- Cardinality ratio

  Participation constraint

  - min - max notation

If you query often → Derived Attribute ×

Relationship type ✓

∵ Fetching is much faster

→ Data Models :

(1) ER Models

(2) UML Class diagram

Entities ⟷ Classes

Relationships ⟷ Lines b/w classes

→ DBMS :

Scripting Language

Embedded Approach

- Interface
- Utilities → Functions

- Cambridge Analytica - 3500 features stored about you

- Centralised DBMS Architechture

  Client - Server Architechture → 3 - Tier

→ Relational Model :

- Relation → Set of rows , i.e. tuple → Ordered set of values (1 tuple : 1 Row)

  Attribute → Column

- Key : Uniquely identifies a row in a table

  → Can be set of attributes (or) attribute

- Schema : Table definition

  $\underbrace{R}_{\text{name of the relation}}(\underbrace{A_1, A_2, \ldots, A_n}_{\text{Attributes}})$

- Domain :

  - All possible column values
  - Datatype constraint

    Format constraint
  - Developed from Attribute name

- Relation State :

  $r(R)$ → Subset of Relation R

  $r(R) = \{t_1, t_2, \ldots, t_n\}$

  $t_i = \langle v_1, v_2, \ldots v_n \rangle$

  v_j → element of domain of A_j

  $r(R) \subset dom(A_1) \times dom(A_2) \times \ldots \times dom(A_n)$

  The tuples are not considered to be ordered.

  The attributes are ordered. ← Self - Describing

23/10

- Value : Atomic

  value v_i of attribute A_i for tuple t ← $t[A_i] = v_i$

- Constraints :

  (1) Inherent (Implicit) : No duplicates

  (2) Schema - based (Explicit) : DB design

  (3) Application - based (Semantic) : By programmers

- Super Key : Attribute / set of attributes that uniquely identify all attributes in a relation

  i.e. for any 2 distinct tuples t_1 & t_2 in $r(R)$: $t_1[SK] = t_2[SK]$

  - Minimal Superkey → Key / Candidate Key
  - Primary Key → Smallest of the candidate Keys
    → Cannot be NULL
  - Foreign Key → Referrential Integrity constraint

27/10

→ Keys

→ Integrity Violation
- Domain constraint
- Key constraint
- Referrential Integrity
- Entity Integrity

ESTRICT

ET NULL

ASCADE

ATION — Owner of Schema

→ Collection of Multiple Schemas

```
SELECT Fname, Mname, Lname
FROM EMPLOYEE
WHERE Dno = (SELECT Dnumber
             FROM DEPARTMENT
             WHERE Dname = "RESEARCH")
```

3/11

AS

LIKE

BETWEEN

ORDER BY

6/11

UPDATE SYNTAX

DELETE SYNTAX

JOINS

- Natural JOIN
- Outer JOIN (Left & Right)
- Inner JOIN